

TVSF Proposal Pipeline — Build Guide (for Paul)

Audience: Paul Jackson (EC / DMVEC shadow-admin for Foundation) and the Foundation team (Kate, Connor). **Goal:** take an Ohio Third Frontier **TVSF** opportunity from the moment it lands on Foundation's cards through a finished, downloadable 3-volume application — the way the EverTrack and HydroSmart winners are structured.

This is a **functional** manual: every screenshot below was captured by driving the live portal as Paul (`e2e/hitl-tvsf-build-guide.spec.ts`), signed in at `pjackson@ecinnovates.com` . If a screen changed, the shot was regenerated — they cannot silently drift.

The canonical TVSF shape (see `docs/TVSF_SPEC.md`) is three volumes:

1. **Narrative** — Abstract (unnumbered) + Questions **1–14**, **7-page** limit, .75" margins, 11 pt Times New Roman. Four tables are mandatory: the **competitor comparison** (Q2), the **pro-forma P&L** (Q6), the **milestone table** (Q11), and the **budget by spend type** (Q12).
2. **Willingness-to-License Letter** — from the IP-owning institution (≤ 1 page). **Required on every TVSF** — it is listed in the compliance matrix with a template example.
3. **ESP Support Letter** — from the Entrepreneurial Services Provider / EC (≤ 1 page).

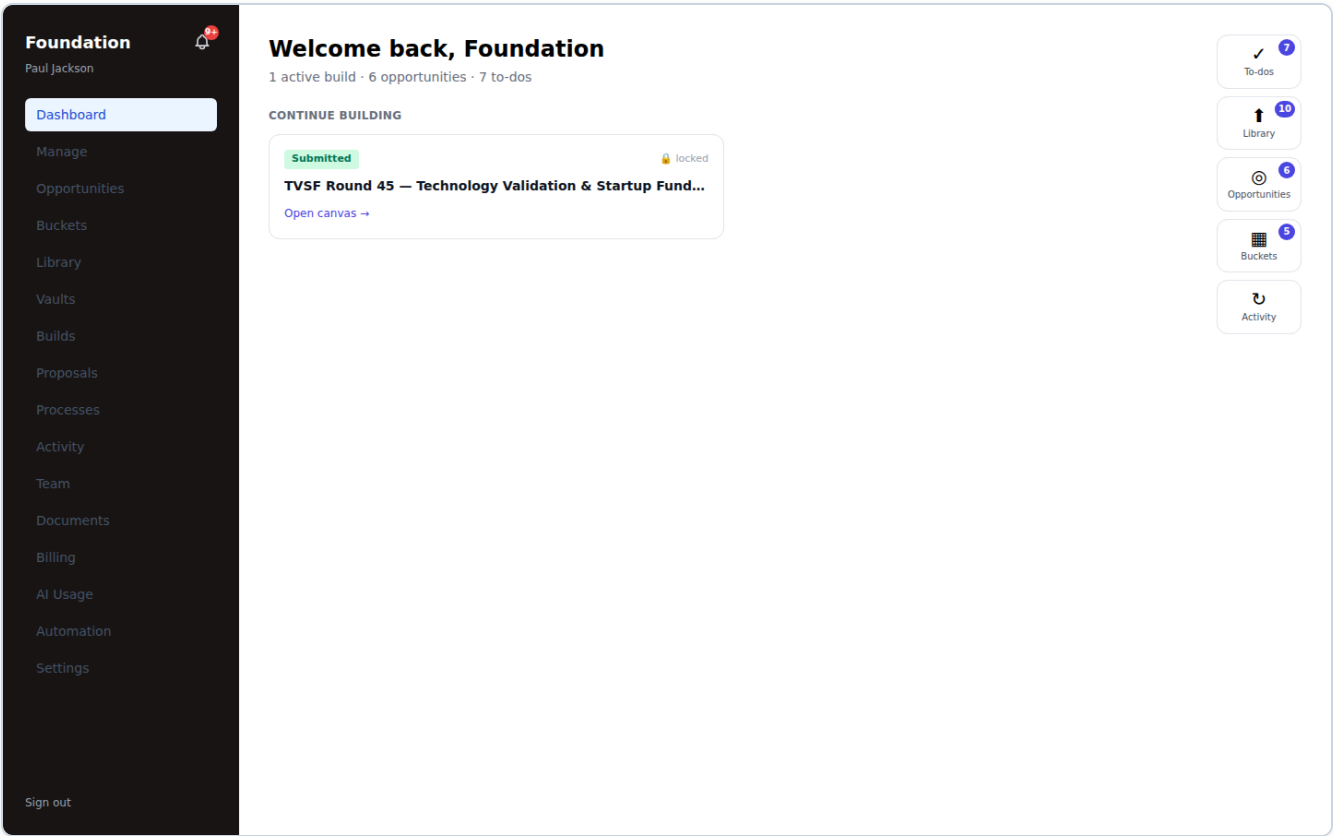
The cast — who touches what

Actor	Identity	Authority
Paul Jackson	<code>pjackson@ecinnovates.com</code> — EC shadow-admin, <i>*external*</i> (no home tenant)	Enters Foundation's account as <code>tenant_admin</code> ; drives discovery, drafting, review, export
Kate Ulepik	<code>kate.ulepic@foundation3dp.com</code> — Foundation, bound <code>tenant_admin</code>	Owens the tenant; edits sections, invites the team
Connor Casey	<code>connor.casey@foundation3dp.com</code> — Foundation <code>tenant_user</code>	Edits assigned sections

Demo password for all three: `DemoPass123!` .

Step 1 — Discovery: from dashboard to the scored Opp Card

Paul lands on Foundation's **dashboard**, the home base for the tenant.



Foundation dashboard

Opportunities are auto-scored into **spotlight buckets** on arrival, so the TVSF round rises to the top of what Foundation should chase.

Foundation

Paul Jackson

Dashboard

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Spotlight Buckets

Your ranking lenses — each ranks the whole pipeline by the criteria you set.

New bucket

Name (e.g. AF Autonomy)

keywords, comma-sep

agencies, comma-sep

program types (SBIR, STTR)

NAICS codes, comma-sep

☐ Include closed opportunities

Create

Rank a bucket to see your pipeline ordered.

Non-dilutive Capital (SBIR/STTR & ...Rank → ×

Materials — Concrete & Low-Carbo...Rank → ×

Construction Technology & HousingRank → ×

Advanced Manufacturing & Autom...Rank → ×

Additive Construction & 3D PrintingRank → ×

Scored buckets

The **Opportunity Cards** surface is the canonical customer view — one denormalized card per activated opportunity. The **TVSF Round 45** card carries its Ohio Third Frontier / DMVEC origin, the Aug 14 2026 close, and the compliance shape the proposal will inherit.

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Opportunity Pipeline

Your live opportunity cards — pin to pull the docs local; ranked by your spotlight buckets.

☐ Include closed [Refresh](#) · 6 cards

Sort: Pinned first

TVSF Round 45 — Technology Validation & Startup Fund (818079)

Ohio Third Frontier · DMVEC (Dayton/Miami Valley Entrepreneurs Center) · tvsf

Closes 8/14/2026

Non-dilutive Capital (SBIR/STTR & State) 83

Materials — Concrete & Low-Carbon Cement 33

Additive Construction & 3D Printing 33

Construction Technology & Housing 33 +1

PinnedUnpinPurchase

Build →

SBIR Phase I — Advanced Manufacturing: Construction Automation & Robotics (903004)

National Science Foundation (NSF) · sbir

Closes 8/28/2026

Advanced Manufacturing & Automation 100

Non-dilutive Capital (SBIR/STTR & State) 75

Construction Technology & Housing 56

Materials — Concrete & Low-Carbon Cement 33 +1

Pin (copy docs)

Build →

SBIR — Low-Carbon Concrete & Cement Materials (903004)

U.S. Department of Energy (DOE) · sbir

Closes 8/20/2026

Materials — Concrete & Low-Carbon Cement 89

Non-dilutive Capital (SBIR/STTR & State) 71

Additive Construction & 3D Printing 43

Construction Technology & Housing 33 +1

Pin (copy docs)

Build →

STTR — Robotics for the Built Environment (903004)

National Science Foundation (NSF) · sttr

Closes 9/19/2026

Non-dilutive Capital (SBIR/STTR & State) 68

Advanced Manufacturing & Automation 53

Construction Technology & Housing 42

Materials — Concrete & Low-Carbon Cement 20 +1

Pin (copy docs)

Build →

Additive Construction of Expeditionary Structures (ACES) (903004)

U.S. Army — ERDC (Engineer Research & Development Center) · sbir

Closes 9/5/2026

Additive Construction & 3D Printing 68

Materials — Concrete & Low-Carbon Cement 64

Non-dilutive Capital (SBIR/STTR & State) 60

Construction Technology & Housing 31 +1

Pin (copy docs)

Build →

SBIR — Additive Construction for Off-World Habitats (903004)

NASA · sbir

Closes 9/12/2026

Non-dilutive Capital (SBIR/STTR & State) 64

Materials — Concrete & Low-Carbon Cement 42

Construction Technology & Housing 31

Additive Construction & 3D Printing 30 +1

Pin (copy docs)

Build →

Opportunity cards

Step 2 — Open the proposal workspace

From **Proposals**, Paul opens the TVSF build. This is the spine of the whole exercise.

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Proposals

1 proposal

TVSF Round 45 — Technology Validation & Startup Fund (818079)

Due Aug 14, 2026 V1 - Submitted

Ohio Third Frontier · DMVEC (Dayton/Miami Valley Entrepreneurs Center) 17 sections Created Aug 1, 2026

Proposals list

The workspace shows the **three volumes** in order — Narrative (Abstract + Q1–14), the Willingness-to-License Letter, and the ESP Support Letter — each section with its draft status, the **Draft** → **Final** gate, and the compliance matrix under the **Compliance** tab. Sections list in true document order (Abstract, 1, 2, ... 14 — never string-sorted "10" before "2").

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TVSF Round 45 — Technology Validation & Startup Fund (818079)

Ohio Third Frontier · DMVEC (Dayton/Miami Valley Entrepreneurs Center) TVSF Due Aug 14, 2026

► Opportunity origin & compliance

► Assign a task

1 Draft — 2 Final

Due: Aug 14, 2026 (13 days)

[Unlock for Edit](#)

Locked (#1) Download available (5 downloaded)

All Sections My Sections Timeline

Artifacts Team & Access Compliance AI & Library Library Seed

17 sections drafted and ready to accept & lock for this stage.

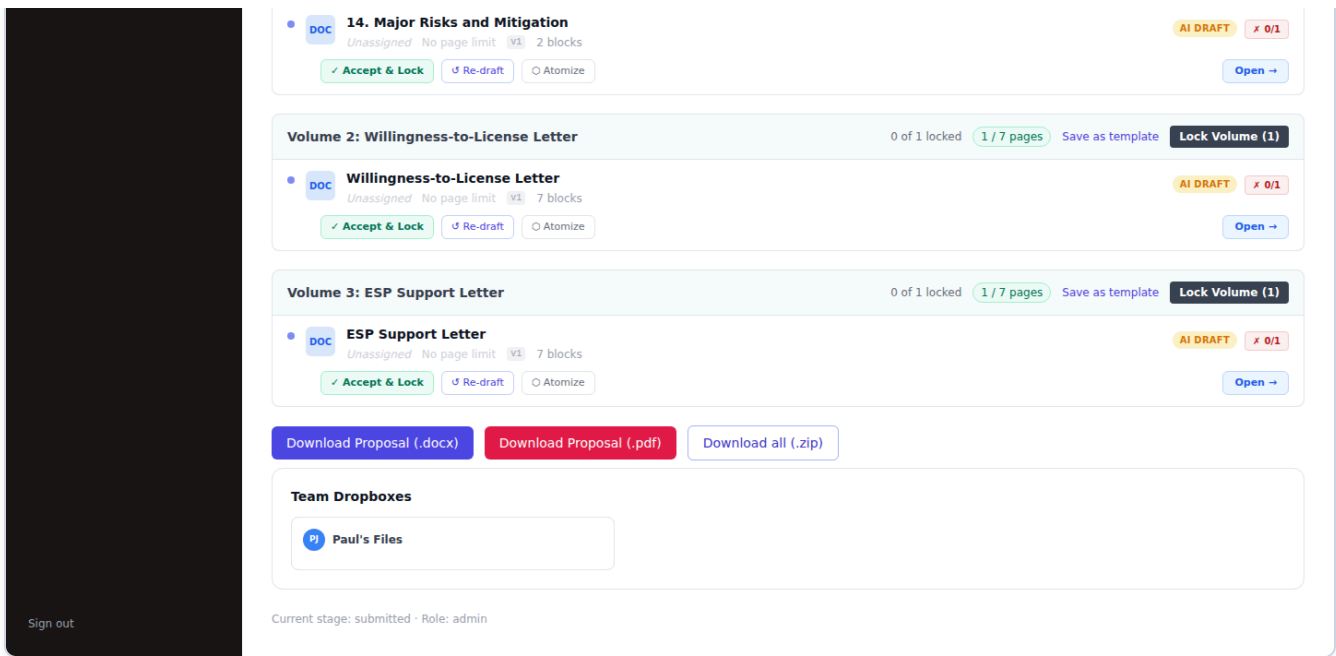
[Accept & Lock All \(17\)](#)[Force advance to V1 →](#)**Volume 1: Narrative**

0 of 15 locked

5 / 7 pages

[Save as template](#)[Lock Volume \(15\)](#)

• DOC	Abstract <i>Unassigned</i> No page limit V1 2 blocks	AI DRAFT x 0/1
	✓ Accept & Lock ↻ Re-draft ○ Atomize	Open →
• DOC	1. Market Opportunity <i>Unassigned</i> No page limit V1 3 blocks	AI DRAFT x 0/1
	✓ Accept & Lock ↻ Re-draft ○ Atomize	Open →
• DOC	2. Overview of Technology/Product <i>Unassigned</i> No page limit V1 3 blocks	AI DRAFT x 0/1
	✓ Accept & Lock ↻ Re-draft ○ Atomize	Open →
• DOC	3. Development Stage and Timeline <i>Unassigned</i> No page limit V1 5 blocks	AI DRAFT x 0/1
	✓ Accept & Lock ↻ Re-draft ○ Atomize	Open →
• DOC	4. Commercialization and Market Entry Strategy <i>Unassigned</i> No page limit V1 3 blocks	AI DRAFT x 0/1
	✓ Accept & Lock ↻ Re-draft ○ Atomize	Open →
• DOC	5. IP Position <i>Unassigned</i> No page limit V1 2 blocks	AI DRAFT x 0/1
	✓ Accept & Lock ↻ Re-draft ○ Atomize	Open →
• DOC	6. Business Model <i>Unassigned</i> No page limit V1 5 blocks	AI DRAFT x 0/1
	✓ Accept & Lock ↻ Re-draft ○ Atomize	Open →
• DOC	7. Current Financial Stage <i>Unassigned</i> No page limit V1 2 blocks	AI DRAFT x 0/1
	✓ Accept & Lock ↻ Re-draft ○ Atomize	Open →
• DOC	8. Economic Impact on State of Ohio <i>Unassigned</i> No page limit V1 2 blocks	AI DRAFT x 0/1
	✓ Accept & Lock ↻ Re-draft ○ Atomize	Open →
• DOC	9. Management Team <i>Unassigned</i> No page limit V1 4 blocks	AI DRAFT x 0/1
	✓ Accept & Lock ↻ Re-draft ○ Atomize	Open →
• DOC	10. ESP Engagement <i>Unassigned</i> No page limit V1 2 blocks	AI DRAFT x 0/1
	✓ Accept & Lock ↻ Re-draft ○ Atomize	Open →
• DOC	11. Project Plan <i>Unassigned</i> No page limit V1 3 blocks	AI DRAFT x 0/1
	✓ Accept & Lock ↻ Re-draft ○ Atomize	Open →
• XLS	12. Budget: Table and Narrative <i>Unassigned</i> No page limit V1 4 blocks	AI DRAFT x 0/1
	✓ Accept & Lock ↻ Re-draft ○ Atomize	Open →
• DOC	13. Next Steps <i>Unassigned</i> No page limit V1 2 blocks	AI DRAFT x 0/1
	✓ Accept & Lock ↻ Re-draft ○ Atomize	Open →



Proposal workspace

At the bottom sit the three **downloads** — `.docx` , `.pdf` , and `.zip` (each volume in its native format). They unlock once the proposal is locked or submitted.

Step 3 — Draft in the canvas (the heart)

Each section opens in the **canvas editor**. The canvas is the document itself — headings, text, tables, figures — rendered exactly as it will export.

Q2 — Overview of Technology/Product carries the mandatory **competitor comparison table** (Foundation vs conventional formwork vs mortar-extrusion printers):

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Narrative

2. Overview of Technology/Product

v1

AI DRAFT

LOCKED

Comp 0/1

read-only

Export

Panel

Overview of Technology/Prod...

ai drafted

Export .docx

Export .pdf

Export .xlsx

Hide panel

read-only

Foundation — Ohio Third Frontier TVSF

2. Overview of Technology/Product

Foundation's system prints a home's foundation-wall geometry directly: a build plan is downloaded to the printer, concrete is pumped into the machine, and the nozzle lays concrete one layer at a time while a printing trolley moves laterally and vertically along runway rails. Two differentiators define the product. First, it uses common, locally sourced concrete rather than an expensive proprietary mortar — the single largest cost and supply-chain constraint on competing printers. Second, an external gate controls material flow at the nozzle, enabling clean starts and stops and consistent layers without the smearing and cold-joint defects that plague mortar-extrusion systems. The net effect eliminates the build-strip-repair-clean formwork cycle, cutting foundation labor from ~336 hours to ~25 and formwork/material cost by 47–65% versus both conventional forming and mortar-based concrete printers.

Performance factor	Foundation	Conventional formwork	Mortar-extrusion printers
Common local concrete (no proprietary mortar)	✓	✓	✗
Clean starts/stops (external gate)	✓	n/a	✗
Eliminates form build-strip-clean cycle	✓	✗	✓
Foundation labor hours (2,100 sq ft)	~25	~336	~120
Formwork + material cost vs conventional	~47–65%	baseline	~10–20%

YOUR TOOLBOX

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comment, revise, and complete (lock) this ...

Compliance & Status

Preview

Export

Compliance

Select

Review

History

DOCUMENT STATUS

Status

ai drafted

Atoms

3

Version

v1

COMPLIANCE

Length

174 / 3675 words

7-page limit · ~3500-word target

Font

Times New Roman 11pt

Margins

0.75" all

CONTENT SOURCES

3 manual

Q2 competitor table

Q3 — Development Stage and Timeline shows a real figure — the 12-month milestone schedule (MS1–MS8), which maps one-for-one to the Q11 milestone table:

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Narrative

3. Development Stage and Timeline

v1

AI DRAFT

LOCKED

Comp 0/1

read-only

Export

Panel

Development Stage and Timeline

ai drafted

Export .docx

Export .pdf

Hide panel

read-only

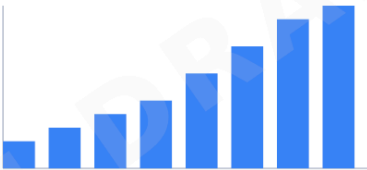
Foundation — Ohio Third Frontier TVSF

3. Development Stage and Timeline

The core printing system is beyond proof-of-concept. Foundation has printed full-height wall segments with the runway-rail trolley and the external-gate nozzle using standard ready-mix concrete, validating the layer geometry and clean start/stop behavior that differentiate the product (TRL 5-6). The customer requirement driving the current MVP came directly from production-builder interviews: a printer that runs on locally batched concrete and produces an inspection-ready footing-to-top-of-wall in a single setup. This TVSF project takes the system to a manufacture-ready configuration and a fully instrumented field demonstration.

The 12-month plan below is summarized as a milestone schedule that matches the Q11 milestones one-for-one; time to first commercial installation is estimated at 14 months from award.

TVSF 12-Month Milestone Schedule (target completion month)



Month	Target Completion Month
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12

Figure 1: TVSF 12-month milestone schedule — target completion month for MSI-MS8, mapping one-for-one to the Q11 Project Plan table.

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comment, revise, and complete (lock) this ...

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Preview

Export

Compliance

Select

Review

History

DOCUMENT STATUS

Status

ai drafted

Atoms

5

Version

v1

COMPLIANCE

Length

135 / 3675 words

7-page limit · ~3500-word target

Font

Times New Roman 11pt

Margins

0.75" all

CONTENT SOURCES

5 manual

Q3 milestone figure

Q6 — Business Model pairs the pro-forma P&L table with a revenue & gross-profit chart, so the recurring lease-plus-build-fee model is both tabulated and visualized:

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← Back to Proposal / Narrative > 6. Business Model

v1 AI DRAFT LOCKED x Comp 0/1

read-only

Export ▾ Panel

Business Model ai drafted

Export .docx

Export .pdf

Export .xlsx

Hide panel

read-only

Foundation — Ohio Third Frontier TVSF

6. Business Model

Foundation generates revenue two ways: printer leases to foundation subcontractors (recurring monthly), and a per-linear-foot build fee tied to each printed foundation (usage-based). Hardware is deliberately not sold outright — the lease-plus-usage model lowers the customer's adoption barrier and gives Foundation a recurring, expanding revenue base as each printer prints more homes. R&D remains focused on print speed, concrete-mix latitude, and multi-story capability. The abbreviated pro-forma below uses the required TVSF categories without modification; figures are directional and in \$1,000s.

(\$1,000s)	2026	2027	2028	2029	2030
Revenues					
Product sales (printers)	0	0	0	0	0
Licensing	0	0	0	0	0
Printer lease + build fee	120	900	3,400	9,200	21,000
Other	0	0	0	0	0
Total revenues	120	900	3,400	9,200	21,000
Production Expenses					
Cost of Goods Sold	90	520	1,600	3,900	8,200
Gross profit	30	380	1,800	5,300	12,800
Other Expenses					
R & D, including IP	600	900	1,300	1,800	2,400
Sales, General & Admin.	350	700	1,400	2,600	4,300
Total other expenses	950	1,600	2,700	4,400	6,700
Net profit	(920)	(1,220)	(900)	900	6,100
Equity Investment	1,500	3,000	0	0	0

Pro-Forma Revenue & Gross Profit, 2026-2030 (\$1,000s)

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comment, revise, and complete (lock) this ...

Compliance & Status

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7-page limit · ~3500-word target

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Margins

0.75" all

CONTENT SOURCES

5 manual

Q6 pro-forma + chart

Foundation team members (Kate, Connor) edit their assigned sections here the same way; edits are versioned and the **color-team / compliance** agents review on advance (see **Agents**, below).

Step 4 — Preview it "as it downloads"

The right-hand toolbox has a **Preview**. It renders the section — or the **whole document** — in the exact print/PDF layout the customer will download, so there are no surprises at export time.

Section preview:

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Section preview

Full-document preview — every volume, in order, clean numbering, header/footer, tables and figures. The banner says it plainly: **"Print / PDF layout — the .docx you download matches this content."**

Preview

This section

Full document

Overview of Technology/Product

Print / PDF layout — the .docx you download matches this content.

Close x

Foundation — Ohio Third Frontier TVSF

Abstract

Residential foundations are still built the way they were a century ago: crews set and strip wooden or aluminum formwork, pour concrete, wait, then repair and clean — a labor-intensive, weather-exposed cycle that adds roughly 336 hours and thousands of dollars to every home. Foundation 3D-prints the foundation wall directly from a downloaded build plan, using common, locally sourced concrete instead of expensive proprietary mortar and an external material-flow gate that produces clean starts, stops, and consistent layers. Foundation is an Ohio company commercializing this printer for the U.S. residential-construction market. This TVSF project will validate a production-ready printer and deliver third-party performance data that de-risks a \$23.7B U.S. formwork opportunity while advancing Ohio's housing-affordability and advanced-manufacturing agendas.

1. Market Opportunity

Every residential concrete foundation in the United States is formed with temporary wooden or aluminum formwork that must be built, set, stripped, repaired, and cleaned for reuse. On a typical 2,100 sq ft home this formwork cycle consumes roughly 336 labor hours and a meaningful share of the foundation budget, and it is a chronic source of schedule slip because it cannot proceed in poor weather. The pain is felt hardest by production homebuilders and foundation subcontractors, who face a persistent skilled-labor shortage and thin margins. Approximately 1.4 million new single-family homes are started in the U.S. each year, essentially all requiring a formed concrete foundation, and the current U.S. spend on residential formwork and foundation labor is estimated at over \$23.7B annually. The addressable pain is the formwork portion of that spend — the build-strip-repair-clean cycle — which is precisely what a direct-printed wall eliminates.

2. Overview of Technology/Product

Foundation's system prints a home's foundation-wall geometry directly: a build plan is downloaded to the printer, concrete is pumped into the machine, and the nozzle lays concrete one layer at a time while a printing trolley moves laterally and vertically along runway rails. Two differentiators define the product. First, it uses common, locally sourced concrete rather than an expensive proprietary mortar — the single largest cost and supply-chain constraint on competing printers. Second, an external gate controls material flow at the nozzle, enabling clean starts and stops and consistent layers without the smearing and cold-joint defects that plague mortar-extrusion systems. The net effect eliminates the build-strip-repair-clean formwork cycle, cutting foundation labor from ~336 hours to ~25 and formwork/material cost by 47–65% versus both conventional forming and mortar-based concrete printers.

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Eliminates form build-strip-clean cycle	✓	✗	✓
Foundation labor hours (2,100 sq ft)	~25	~336	~120
Formwork + material cost vs conventional	~47–65%	baseline	~10–20%

Sign out

Full document preview

Step 5 — Build from Foundation's library atoms

Reusable, taxonomy-tagged **library atoms** (uploaded material, atomized) are the raw stock the drafter and the team pull from. Everything is visibility-enforced to Foundation.

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Library Atoms

+ Create canvas

Shred your own documents by hand — deconstruct into objects, keep only the content you like, tag it against the one taxonomy, and it's sized to drop into section molds. Primitives (a bio, a figure) combine into groups (a Team) with lineage back to their source. Or create a canvas from scratch — a foundation artifact that decomposes into those same atoms.

Browse library (0)

Search title / summary...

AllFoundationsSectionsGroupsAtoms

Nothing matches these filters.

Reuse a past proposal

templify its structure → regenerate fresh drafts

Turn a past proposal you've uploaded into a reusable template — its winning structure, content stripped — then start a fresh draft from it, filled from your library.

No past proposals yet. Upload a prior proposal package below (it's atomized into your library), then come back here to templify it.

Library

Upload package

Atomize

Capture

Search title / summary...

all grains

any status

All atoms

Refresh

Select all 10 atoms

☐

primitive

Budget basis — TVSF \$200k by spend type

47 words

used 13x

UPLOADED

admin

approved

archive

Personnel \$40k, Equipment \$120k, Supplies \$10k, Purchased Services \$30k = \$200k. Bulk = equipment (COTS+parts) + Converge services.

kind:budget_data

program:tsvf

vol:cost

☐

primitive

Team / management

54 words

used 13x

UPLOADED

admin

approved

archive

Kate Uleplic CEO, Conor Atkins COO, Connor Casey CFO, Will Curley CTO. Gaps: legal advisor, fundraiser, technician, homebuilder. EC-mentored (Paul Jackson).

kind:bio

program:tsvf

vol:team

☐

primitive

Project plan & milestones

55 words

used 13x

UPLOADED

admin

approved

archive

Hire (Mo1-2 \$10k) → R&D w/ Converge (Mo3-4 \$190k) → beta prototype (Mo6-8) → third-party validation (Mo9-14 \$30k) → first printer build (Mo11-14 \$300k).

kind:narrative

program:tsvf

vol:plan

☐

primitive

Business model & financials

74 words

used 13x

UPLOADED

admin

approved

archive

JV per project; charge per cubic yard; ≈\$20k/job; ≈37.5% contribution margin. Raise: \$200k TVSF + \$150k SBIR + \$1.8M+ equity; pre-seed.

kind:narrative

program:tsvf

vol:business

☐

primitive

Intellectual property — 2 issued patents

42 words

used 13x

UPLOADED

admin

approved

archive

U.S. Patents 11,273,574 (Scalable 3D Printing Apparatus) and 10,307,959 B2 (Concrete Delivery System).

kind:narrative

program:tsvf

vol:ip

☐

primitive

Development stage & timeline (TRL 6-7)

56 words

UPLOADED

admin

approved

archive

TRL 6-7; beta prototype in progress; 14-month Gantt through third-party validation + first printer build. MVP: rapid deploy, reduced labor, structural reliability, local ...

kind:narrative

program:tsvf

vol:technical

☐

primitive

Competitive analysis — vs Traditional / Vitruvian / COBOD

53 words

UPLOADED

admin

approved

archive

Foundation is the only option combining local-material accessibility + lowest project cost with rapid, scalable, repeatable deployment.

kind:narrative

program:tsvf

vol:competitive

☐

primitive

Market — TAM/SAM/SOM + formwork wedge

62 words

UPLOADED

admin

approved

archive

TAM \$410B (2026 housing), SAM \$23.7B (SF-newbuild formwork), SOM \$27.4M; ~500k supply gap; foundation 3D printing cuts total home cost ~4%.

kind:narrative

program:tsvf

vol:market

☐

primitive

Technology overview — print process + differentiators

87 words

UPLOADED

admin

approved

archive

Downloads plan → pumps concrete → lays layers; trolley moves lat/vert on rails; electric. Uses common local concrete (not mortar); external nozzle gate.

kind:narrative

program:tsvf

vol:technical

☐

primitive

Foundation — company overview

59 words

UPLOADED

admin

approved

archive

Ohio pre-seed venture that 3D-prints residential foundation formwork; ~47% formwork cost + ~11 days saved on a 2,100 sq ft home.

kind:narrative

program:tsvf

vol:overview

Library atoms

Step 6 — The Foundation team

Paul and Kate manage the team from **Team** — inviting Foundation members and scoping their access.

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Sign out

Team

4 members

Invite Team Member

Email *

colleague@company.com

Name

Full name

Role

Contributor

Send Invite

Team Members

Name	Email	Role	Status	Last Login	Actions
Kate Ulepik	kate.ulepic@foundation3dp.com	Admin	Active	Aug 2, 2026	Admin Deactivate
Conor Atkins	conor.atkins@foundation3dp.com	Contributor	Active	Never	Contributor Deactivate
Connor Casey	connor.casey@foundation3dp.com	Contributor	Active	Aug 2, 2026	Contributor Deactivate
Will Curley	will.curley@foundation3dp.com	Contributor	Active	Never	Contributor Deactivate

Proposal Collaborators

No external collaborators on active proposals.

Foundation team

Step 7 — Export the finished application

Back in the workspace, the three download buttons produce:

- **Download Proposal (.docx)** — one assembled Word document, all sections combined, tables and figures embedded.
- **Download Proposal (.pdf)** — one print-fidelity PDF (Chromium render): repeating header/footer, real page numbers, tables, and the inline figures at full resolution.
- **Download all (.zip)** — each volume exported in its *native* format (the narrative as `.docx`, the budget grain as `.xlsx`, etc.), bundled — lossless for a mixed proposal.

The delivered Foundation application is **6 pages**: Abstract + Q1–14 with all four mandatory tables and both figures, followed by the Willingness-to-License and ESP Support letters — numbering clean end to end.

The brain & nervous system — agents and automation

The build isn't manual busywork. The pipeline's agent workforce plugs into the workflow engine (`docs/AGENT_WORKFORCE.md`, `docs/AUTOMATION_SPINE_MAP.md`):

- **section_drafter** drafts each section from the library atoms on release/provision (`draft_v0` → `markdown_to_canvas` → `publish_section_draft`) — that's how the Q1–14 drafts and the letters arrive as `ai_drafted` .
- **compliance_reviewer** runs inline (`ai/compliance`) against the matrix — including the Willingness-to-License and ESP letter requirements.
- **color_team_reviewer** reviews on advance via the `agent_task_queue` .
- **librarian** atomizes uploads into `library_atoms` ; **scoring_strategist** feeds the buckets.

Every agent is **advisory** → **guardrail** → **land-or-review**: it never auto-writes the business tables, tenant content is injection-fenced, and a run is tenant-bound. Oversight lives at `/admin/agents` (Agent Workforce).

Appendix — the TVSF compliance checklist

#	Requirement	Where
—	Abstract (unnumbered, excluded from the 7-page count)	Narrative
1–14	The 14 numbered questions	Narrative
Q2	Competitor comparison table	Narrative
Q6	Pro-forma P&L table (+ revenue chart)	Narrative
Q11	Milestone table (+ Q3 schedule figure)	Narrative
Q12	Budget by spend type table	Narrative
—	Willingness-to-License letter (IP owner, ≤ 1 page)	Volume 2
—	ESP support letter (the EC, ≤ 1 page)	Volume 3
—	Format: 7-page narrative, .75" margins, 11 pt Times New Roman, 12 pt spacing	all

To refresh the demo proposal to this canonical shape at any time: `DATABASE_URL=... node scripts/rebuild-tvsf.mjs` .